

Pseudo-visibility: what the information advantage actually proves

A correction to P1a and a restricted proof of P1b (note 09, section 1)

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Provenance. Produced by OpenAI Codex during an external audit of this repository (27 September 2026), and checked by Anthropic Claude before inclusion. Hand proofs; no independent human review; priority not established.

Source. S. A. Alexander, *Pseudo-visibility: a game mechanic involving willful ignorance*, FLAIRS-35, 2022 ([doi:10.32473/flairs.v35i.130652](https://doi.org/10.32473/flairs.v35i.130652)): Definition 2, the discussion after Lemma 4, and Definition 10.

1. A frozen random seed does not imply independent draws

The source suggests retaining a policy’s random-generator state between action queries and advancing it during training only. This makes a policy a fixed function of observation and the retained seed during a step.

Counterexample to this repository’s inference: retain a fair bit Z for the whole step, and let the plain policy output Z at **every** observation. At both the erased and visible observations its marginal action distribution is $(1/2, 1/2)$, but its two outputs agree with probability 1. There is no mismatch penalty. This repository’s proposed $q(m-1)\delta$ penalty floor would give $q/2$ for $m=2$, $\delta=1/2$. Its independence premise is therefore an extra model assumption, not a consequence of the source’s random-generator advice.

This repository’s script explicitly implements a different experiment: an isolated query RNG supplies the query, a separate RNG supplies the actual action, and the annotated arm reuses the annotation as the environment’s query answer. That is a usable private-query protocol, but it is not evidence for an unconditional capacity advantage in the frozen-seed source protocol.

2. Exact value of a genuinely private counterfactual annotation

Fix one context. Suppose C has a fixed distribution p on a finite action set, is hidden from the plain policy and revealed to the annotated policy, and does not change distribution when the visible policy is changed. Let $R(a)$ be expected base reward and $q \geq 0$. Assume R does not additionally depend on C .

Theorem. The best plain and annotated one-step values are respectively

$$V_{\text{plain}} = \max_a \{R(a) - q(1 - p_a)\}, \quad V_{\text{ann}} = \sum_c p_c \max_a \{R(a) - q\mathbf{1}[a \neq c]\}.$$

Consequently $V_{ann} \geq V_{plain}$, but strictness is not automatic.

Proof. Any unannotated random action is a convex combination of the displayed action values, so a pure maximizing action suffices. Once $C=c$ is observed, maximize separately in that conditional problem. The annotated policy can always ignore C , proving the inequality. ■

If R is constant, then copying C is optimal and the gain is exactly $q(1 - \max p_a)$, at least $q(m-1)\delta$ when every $p_a \geq \delta$. **This is a valid restricted version of P1a.** It depends on private information and a matched base-reward condition, not self-reference alone.

For an explicit failure of general strictness, take two actions, uniform C , $q=1/4$, and $R(0)=1$, $R(1)=0$. Action 0 is optimal even after either annotation, so both optimal values are $7/8$. Always copying C instead has value $1/2$. Avoiding a penalty can cost more in base reward than it saves.

This theorem holds the erased-action distribution fixed. Optimizing the entire self-referential policy can also change that distribution; that is a separate optimization problem and must not be smuggled into this conditional comparison.

3. A precise restricted version of P1b

Consider the finite contextual **one-step bandit** implemented by note 09's greedy-query tabular experiment. Contexts have positive sampling probability; both erased and visible flags occur with positive probability. Returns have stationary means $R(x,y,a)$ and bounded conditional second moments. Use tabular sample-average updates, and visit each action infinitely often at each state that continues to occur. Queries use the current greedy erased action.

Assume, for every context x :

1. $R(x,0,\cdot)$ has a unique maximizer $a_0(x)$.
2. $R(x,1,a) - q \mathbb{1}[a \neq a_0(x)]$ has a unique maximizer $a_1(x)$.

Theorem. The final greedy policies of the plain and annotated tabular learners converge almost surely to the same pair $(a_0(x), a_1(x))$.

Proof. Erased-state returns are stationary and never include the visibility penalty. Infinite visits and the sample-average law of large numbers give convergence of every erased action estimate to $R(x,0,a)$. Uniqueness and the finiteness of the context/action sets imply that all greedy erased actions eventually stabilize simultaneously at $a_0(x)$.

Thereafter, visible-state rewards have stationary means $R(x,1,a) - q \mathbb{1}[a \neq a_0(x)]$. For the annotated learner only the state $(x,1,a_0(x))$ continues to occur; the other annotation states can be disregarded when evaluating the eventual policy. Infinite action visits at the remaining states and sample-average updates remove the finite pre-stabilization data. The visible estimates converge to the same means in both learners. Their unique maximizers give eventual greedy choice $a_1(x)$. ■

For this repository's example, $p=1/2$, $q=2$, $R_0=(1,0.9)$, $R_1=(0,1)$, this limit is $(0,0)$ with value 0.50; the joint optimum is $(1,1)$ with value 0.95. Both uniqueness conditions hold. Thus the finite experiment's observed gap has a hand proof under the assumptions just stated. It is a gap in the **evaluation of the limiting greedy policies**; persistent exploration changes training returns.

The theorem does not cover constant learning rates, tied maxima, shared linear features, multi-step MDPs, general policy-dependent environments, or a sampled query. This repository's finite time points alone establish neither asymptotic convergence nor a learning-rate theorem in those settings. Its $O(1)$ versus $\Omega(K)$ conjecture also needs a specified error tolerance, confidence, context distribution and feature access before it is a sample-complexity statement.

Finite exact controls for the value formula and counterexamples are recorded in the audit controls (`../audit-2026-09-27/controls.json`). No new training campaign or claim of external novelty is involved.